ALICE L WALTON SCHOOL OF MEDICINE

Digestive Motility

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Learning Objectives (5)

After completing this brick, you will be able to:

1. Distinguish tonic from phasic smooth muscle contractions, identify where each pattern predominates along the GI tract, and explain how sphincter tone regulates flow between compartments.
2. Explain how ICC-generated slow wave potentials set the basal contraction frequency of GI smooth muscle and gate action potential firing, and distinguish this from the mechanisms of skeletal mus
3. Distinguish peristaltic from segmental contractions in terms of neural coordination, directionality, and functional role in propulsion vs. mixing.
4. Predict how disruption of ICC pacemaker activity alters the frequency and coordination of GI smooth muscle contractions.
5. Explain the phases and motilin-dependent regulation of the migrating motor complex, and predict the consequences of its absence on intestinal function between meals.

CASE CONNECTION

LT, a 37-year-old male, presents at the gastroenterology clinic with a 4-month history of intermittent constipation and unexplained bloating. He reports infrequent bowel movements, a sense of incomplete evacuation, and postprandial abdominal distension. He denies any recent dietary changes, travel, or antibiotic use.

What are the key patterns of smooth muscle contractions along the GI tract, and how might disruptions in coordination affect symptoms like constipation and bloating? Consider your answers as you read, and we'll revisit LT at the end of the brick.

GO TO CONCLUSION ↓

Tonic vs. Phasic Smooth Muscle Contractions (LO1)

GI smooth muscle employs two fundamentally different contraction strategies—tonic and phasic—each suited to distinct functional demands along the digestive tract.

- **Tonic contractions** are sustained contractions that persist for minutes to hours without significant relaxation; rather than relying on rhythmic electrical events, they maintain steady muscle tension ideally suited to structures that must remain closed or constricted.
- **Phasic contractions** are rhythmic, intermittent contractions driven by discrete electrical events, well-suited for propelling or mixing luminal contents.

GI Tract Locations

The distribution of tonic and phasic contractions along the GI tract reflects their functional roles. Phasic contractions dominate in regions responsible for active propulsion and mixing, while tonic contractions are concentrated at structural boundaries that regulate compartmental flow.

Table 1: GI Contraction Type by Location

Contraction Type	Primary Locations	Functional Role
Tonic	Sphincters (LES, pyloric, ileocecal valve, internal anal sphincter), orad stomach (fundus)	Barrier maintenance, reservoir function
Phasic	Esophageal body (smooth muscle portion), caudad stomach (antrum/corpus), small intestine, colon	Propulsion and mixing

Sphincter Tone Regulation

Sphincters throughout the GI tract rely on tonic contractions to maintain a baseline state of closure, preventing retrograde flow and controlling the timing of luminal transit. Key sphincters include the lower esophageal sphincter (LES), pyloric sphincter, ileocecal valve, and internal anal sphincter.

Sphincter tone is dynamically regulated by neural and hormonal inputs. For example, the LES relaxes in response to swallowing via vagal inhibitory neurons that release nitric oxide and vasoactive intestinal peptide, allowing food to pass into the stomach. Failure of this relaxation—as in achalasia—illustrates how disrupted tonic regulation can have significant clinical consequences. Conversely, inappropriately frequent transient relaxations of the LES underlie gastroesophageal reflux disease (GERD), demonstrating that both excessive and insufficient tone are pathologically relevant.

CLINICAL CORRELATION

ACHALASIA AND GERD: OPPOSITE FAILURES OF LES TONE

Achalasia results from loss of inhibitory NANC neurons in the myenteric plexus that normally release NO and VIP to relax the LES. Without this signal, the LES cannot relax during swallowing, producing progressive dysphagia for both solids and liquids. Classic findings: incomplete LES relaxation and elevated resting pressure on manometry; "bird's beak" tapering on barium swallow.

GERD, in contrast, results from abnormally frequent transient LES relaxations (TLESRs), not from chronically low resting tone, allowing acid reflux without a swallowing trigger. ures of LES tone: failed relaxation versus inappropriate relaxation.

What are the key distinguishing features of tonic vs. phasic smooth muscle contractions in the GI tract?

ICC–GENERATED SLOW WAVE POTENTIALS (LO2)

Unlike skeletal muscle, which only contracts when directly stimulated by a motor neuron, GI smooth muscle exhibits spontaneous rhythmic electrical activity. This intrinsic rhythmicity is generated by specialized pacemaker cells called the interstitial cells of Cajal (ICC), which set the tempo of phasic contractions throughout the gut.

Slow Wave Generation by ICCs

GI smooth muscle is smooth muscle—cells are electrically coupled through gap junctions so that depolarization in one cell spreads to its neighbors, allowing the muscle layer to contract as a coordinated sheet. This property is what enables ICC-generated slow waves to propagate across the muscle wall. **Multi-unit smooth muscle**, by contrast, lacks extensive gap junction coupling and requires direct innervation of individual fibers — a pattern found in structures such as the iris and some blood vessel walls, but not in the gut wall.

ICCs are located within the myenteric plexus and the smooth muscle layers of the GI wall. They generate rhythmic oscillations in membrane potential known as slow waves—also called the basic electrical rhythm (BER). Slow waves arise from cyclical changes in ion conductance, particularly involving calcium (Ca^{2+}) and chloride (Cl^-) channels, which cause the membrane potential to depolarize and repolarize in a regular pattern.

ICCs are electrically coupled to adjacent smooth muscle cells via gap junctions, allowing slow waves to propagate across the muscle layer. Importantly, slow waves alone may produce only weak contractions in some regions (e.g., the gastric corpus); strong phasic contractions require superimposed action potentials (APs). Slow waves bring the membrane potential closer to the threshold for AP firing, and this gating function means that slow waves set the ceiling contraction frequency, but whether a strong contraction actually occurs depends on additional excitatory inputs.

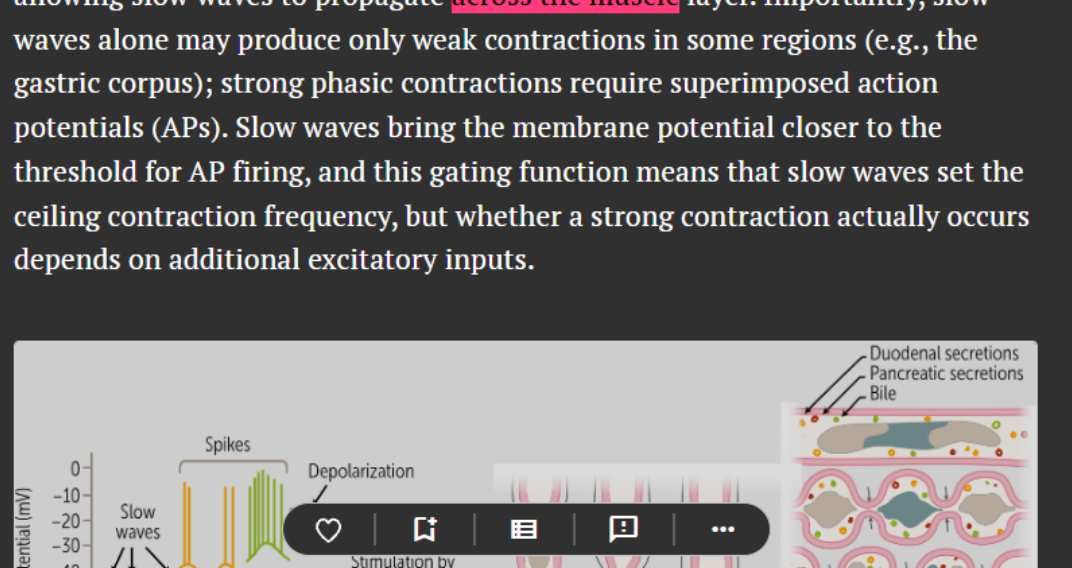

Figure 1: The regulation of phasic electrical activity drives smooth muscle contraction. Stretch, ACh, and parasympathetic signaling depolarize muscle cells, increasing contraction frequency and strength; sympathetic and catecholamine signaling hyperpolarize muscle cells, decreasing contractions. Contractions produce peristalsis and segmentation.

Basal Contraction Frequency

The frequency of slow waves—and therefore the ceiling contraction frequency—varies by region. In the stomach, ICCs generate approximately 3 slow waves per minute, setting the pace for gastric peristalsis. In the duodenum, the frequency rises to about 12 per minute, while in the ileum it falls to approximately 8 per minute. This frequency gradient has functional significance: within the small intestine, it contributes to aborad coordination of peristaltic contractions, since faster proximal pacemakers tend to drive the rhythm distally. Coordinated propulsion also depends on ENS reflex arcs (covered in Neural Control of the GI System).

Mechanism Comparison

In skeletal muscle, there is no intrinsic pacemaker activity—contraction requires a neuromuscular junction signal from an alpha motor neuron, and the muscle is electrically silent without it. Cardiac muscle, like GI smooth muscle, has intrinsic pacemal liac pacemaker potentials are driven primarily by the "funny current" (I_f), a mixed Na^+/K^+ inward current, and result in APs that directly trigger contraction with each cycle.

GI smooth muscle occupies a middle ground: ICCs generate slow waves that modulate—but do not mandate—contraction. The slow wave gates AP firing, and APs are required for contraction. This arrangement allows the enteric nervous system (ENS) and hormonal signals to modulate whether a given slow wave results in a contraction, providing far greater modulatory flexibility. Excitatory inputs, such as acetylcholine acting on muscarinic receptors, depolarize the smooth muscle membrane, bringing the peak of each slow wave closer to AP threshold. Inhibitory inputs such as nitric oxide and vasoactive intestinal polypeptide hyperpolarize the membrane, moving the slow wave peak further from threshold. This gives the ENS direct control over whether pacemaker activity translates into mechanical contraction.

ACTIVE LEARNING QUESTION

Which statement best distinguishes GI smooth muscle slow waves from cardiac pacemaker potentials?

- A. ○ Generated by motor neurons
- B. ○ Identical to action potentials
- C. ○ Do not always trigger contraction
- D. ○ Origi
- E. ○ Absent in the duodenum

Explanation (requires correct answer)

PERISTALTIC VS. SEGMENTAL CONTRACTIONS (LO3)

Once the electrical framework for contraction is established, the ENS orchestrates two distinct patterns of phasic contraction: peristalsis and segmentation. These patterns serve complementary but different purposes—one moves content forward, the other ensures thorough mixing.

Neural Coordination

Peristalsis is coordinated by the peristaltic reflex, also called the law of the intestine. When the intestinal wall is distended by luminal content, the ENS activates an ascending excitatory pathway that contracts the circular muscle oral to the bolus and a descending inhibitory pathway that relaxes the circular muscle anal to the bolus. This coordinated contraction-relaxation wave propels content aborally. More information about neurotransmitters and mechanisms mediating each limb are detailed in Neural Control of the GI System.

Segmental contractions, by contrast, involve localized, non-propagating contractions of circular smooth muscle at multiple points along the intestine simultaneously. These contractions are not coordinated in a directional sequence; instead, th gments that repeatedly contract and relax in an alternating pattern. The ENS coordinates this activity locally, without the ascending/descending reflex arc required for peristalsis.

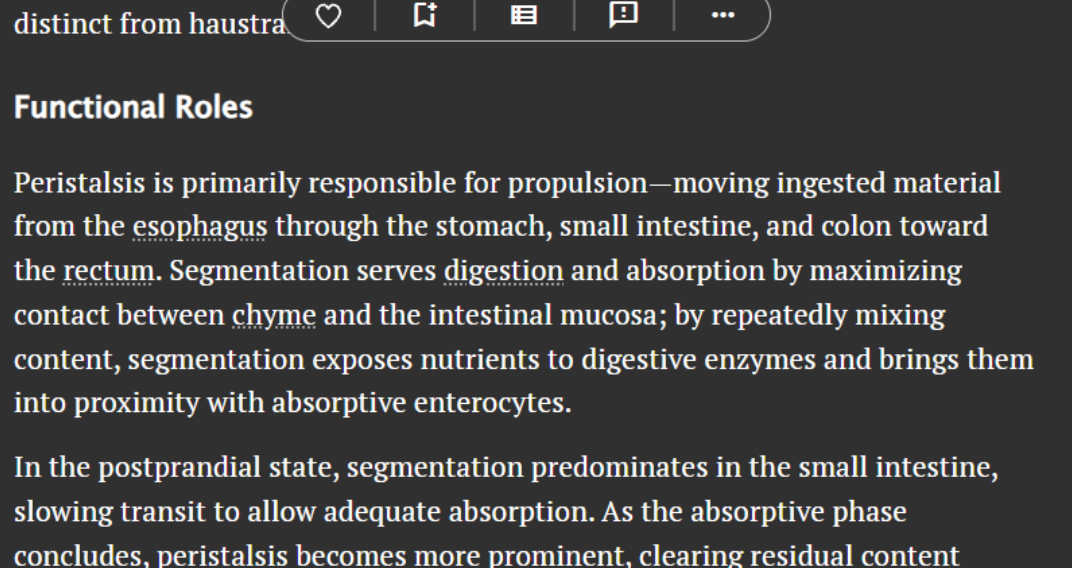
Small intestine
Mucosa
Longitudinal layer of smooth muscle
Circular layer of smooth muscle
Longitudinal layer of smooth muscle
Circular layer of smooth muscle

Figure 2: Arrangement of GI smooth muscle and mucosa.

In the colon, the equivalent pattern is **haustral contractions** — localized, non-propagating contractions of the circular muscle within the sacculations (haustra) formed by the teniae coli and circular muscle tone. Like small intestinal segmentation, haustral contractions have no net directionality; they mix and expose colonic contents to the mucosa, facilitating water absorption rather than propulsion. Mass movements — infrequent, high-amplitude propagating contractions — serve the propulsive role in the colon and are distinct from haustra

Functional Roles

Peristalsis is primarily responsible for propulsion—moving ingested material from the esophagus through the stomach, small intestine, and colon toward the rectum. Segmentation serves digestion and absorption by maximizing contact between chyme and the intestinal mucosa; by repeatedly mixing content, segmentation exposes nutrients to digestive enzymes and brings them into proximity with absorptive enterocytes.

In the postprandial state, segmentation predominates in the small intestine, slowing transit to allow adequate absorption. As the absorptive phase concludes, peristalsis becomes more prominent, clearing residual content distally. This dynamic balance is tightly regulated by the ENS and modulated by hormonal signals such as serotonin (5-HT), which enhances both peristalsis and segmentation.

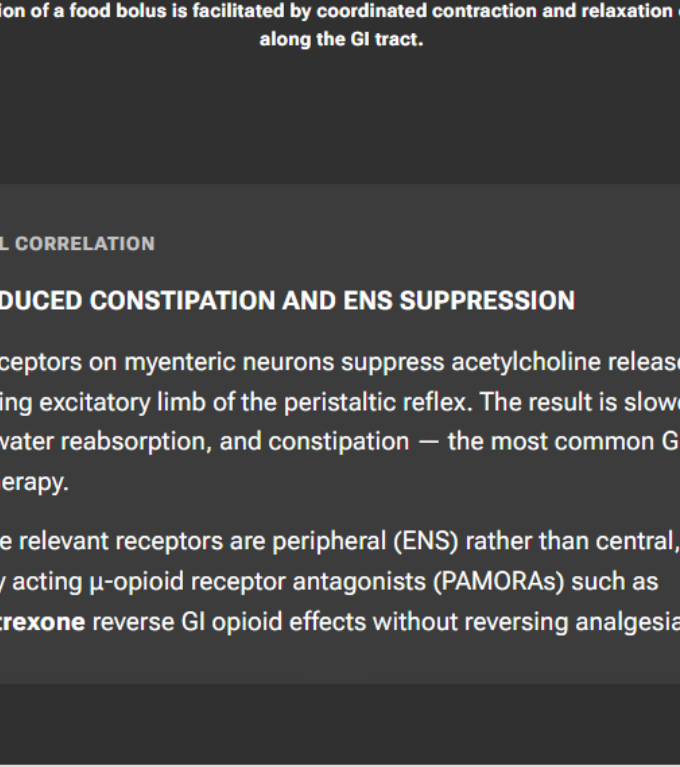

Figure 3: Propulsion of a food bolus is facilitated by coordinated contraction and relaxation of smooth muscle along the GI tract.

CLINICAL CORRELATION

OPIOID-INDUCED CONSTIPATION AND ENS SUPPRESSION

μ-opioid receptors on myenteric neurons suppress acetylcholine release, blunting the ascending excitatory limb of the peristaltic reflex. The result is slowed transit, increased water reabsorption, and constipation — the most common GI side effect of opioid therapy.

Because the relevant receptors are peripheral (ENS) rather than central, peripherally acting μ-opioid receptor antagonists (PAMORAs) such as **methylnaltrexone** reverse GI opioid effects without reversing analgesia.

ACTIVE LEARNING QUESTION

A patient undergoes small bowel resection. Postoperatively, nutrients pass too quickly through the remaining intestine. Which motility pattern is most likely impaired?

- A. ○ Peristalsis
- B. ○ Segmentation
- C. ○ Migrating motor complex
- D. ○ Tonic contraction
- E. ○ Slow wave generation

Explanation (requires correct answer)

DISRUPTION OF ICC PACEMAKER ACTIVITY (LO4)

Because ICCs serve as the primary pacemakers of GI smooth muscle, any disruption to their number, distribution, or function has direct consequences for motility.

Altered Frequency

When ICC density is reduced or ICC function is impaired, slow wave generation becomes irregular or slows in frequency—directly reducing the rate of phasic contractions. In the stomach, for example, loss of ICC activity contributes to gastroparesis — delayed gastric emptying in the absence of mechanical obstruction (see Clinical Correlation below).

Conversely, abnormal ICC activity can also produce ectopic pacemaker foci—regions where slow waves originate outside the normal pacemaker zone. This can generate dysrhythmias in the slow wave pattern, analogous to cardiac arrhythmias, resulting in uncoordinated or retrograde contractions. These slow wave dysrhythmias have clinical names analogous to their cardiac counterparts: **tachygastria** refers to abnormally rapid gastric slow waves (>4 cycles/min), while **bradygastria** refers to an abnormally slow rate (<2 cycles/min). Both disrupt the coordinated antral contractions required for effective gastric emptying and can be detected non-invasively using electrogastrography (EGG).

Coordination Effects

Beyond frequency, ICCs are essential for spatial coordination of contractions. The propagation of slow waves through ICC networks ensures that contractions occur in an organized, sequential manner along the gut wall. When ICC networks are disrupted, as seen in diabetic gastroparesis, the coordinated propagation of slow waves breaks down. The result is segmental or uncoordinated contractions that fail to generate effective propulsion.

Patients with slow transit constipation and intestinal pseudo-obstruction often have significantly reduced ICC density in affected bowel segments, confirming that ICC integrity governs both the rate and directional coordination of phasic contractions.

CLINICAL CORRELATION

DIABETIC GASTROP

Chronic hyperglycemia damages ICCs in the gastric antrum, disrupting slow wave generation and producing delayed gastric emptying without mechanical obstruction. Classic presentation: postprandial nausea, vomiting, early satiety, and bloating in a patient with long-standing diabetes.

First-line pharmacologic management is **metoclopramide**, a D_2 receptor antagonist with prokinetic action that accelerates gastric emptying.

What are the main consequences of ICC pacemaker dysfunction on GI motility?

THE MIGRATING MOTOR COMPLEX AND MOTILIN (LO5)

The small intestine operates in two distinct motor programs depending on digestive state.

During feeding, segr g mucosal contact for digestion and absorption — with peristalsis increasing as the absorptive phase concludes to clear residual content distally.

Between meals, when the postprandial pattern of segmentation and peristalsis subsides, the GI tract does not simply go quiet. Instead, a distinct cyclical pattern of motor activity called the migrating motor complex (MMC) sweeps through the stomach and small intestine. The MMC serves as the gut's "housekeeper," clearing residual content and preventing bacterial overgrowth during the interdigestive period.

During fasting, the segmentation/peristalsis pattern is replaced by the migrating motor complex (MMC), a cyclical housekeeping program that sweeps the lumen clean between meals. The transition between these programs is regulated by the presence or absence of nutrients in the lumen: feeding suppresses the MMC (via suppression of motilin secretion), while the return to fasting restores it.

Phases of Migrating Motor Complex

The MMC cycles approximately every 90–120 minutes during fasting and consists of four distinct phases:

1. Phase I (quiescence): Minimal or no contractile activity; lasts approximately 40–60 minutes
2. Phase II (irregular activity): Intermittent, irregular contractions begin; lasts approximately 20–30 minutes
3. Phase III (activity front): Intense, regular contractions at maximum slow wave frequency propagate aborally; lasts 5–10 minutes and is the defining feature of the MMC
4. Phase IV (transiti transitioning back to Phase I

Phase III is the most clinically significant phase. It generates a powerful, propagating wave of contractions that moves from the stomach to the terminal ileum, effectively sweeping the lumen clean of undigested debris, desquamated cells, and bacteria. The characteristic "stomach growling" (borborygmi) heard during fasting is often the audible result of Phase III activity.

Role of Motilin

The MMC is primarily regulated by motilin, a 22-amino acid peptide hormone secreted by Mo cells (enteroendocrine cells) in the duodenum and jejunum. Motilin is released in a cyclical pattern during fasting, with peak secretion coinciding with the onset of Phase III. It acts on motilin receptors on smooth muscle cells and enteric neurons to initiate the intense contractile activity of Phase III. Conditions that damage ENS integrity, such as diabetic autonomic neuropathy or scleroderma, disrupt the regular MMC rhythm, a loss of interdigestive housekeeping that predisposes to bacterial overgrowth.

Motilin secretion is suppressed by feeding, particularly by the presence of fat and glucose in the duodenum, which explains why the MMC is interrupted when a meal is consumed. The ENS also plays a modulatory role, and the vagus nerve contributes to MMC coordination, though motilin remains the dominant hormonal driver. Erythromycin, a macrolide antibiotic, is a well-known motilin receptor agonist and is used clinically as a prokinetic agent to stimulate gastric emptying in patients with gastroparesis.

Impact of MMC Absence

When the MMC is ab ilin deficiency, vagotomy, or conditions affecting the ENS—the interdigestive housekeeping function is lost. Without the regular Phase III sweep, residual food particles, secretions, and bacteria accumulate in the small intestine. The most significant consequence is small intestinal bacterial overgrowth (SIBO), in which bacteria proliferate abnormally in the proximal small bowel, leading to bloating, flatulence, malabsorption, and diarrhea.

MMC disruption is also associated with nausea, abdominal discomfort, and impaired transit between meals. Conditions such as scleroderma, diabetic autonomic neuropathy, and post-surgical states can all impair MMC function—a pattern directly relevant to the interdigestive symptoms seen in LT's case.

CLINICAL REASONING

MMC DISRUPTION, SIBO, AND THE INTERDIGESTIVE PERIOD

When the MMC is impaired, Phase III sweeps fail and bacteria accumulate in the proximal small bowel. Bacterial fermentation of carbohydrates produces hydrogen and methane gas, causing bloating, flatulence, and diarrhea. Bile acid deconjugation by overgrown bacteria contributes to fat malabsorption, and bacterial consumption of dietary B12 produces deficiency.

Diagnosis: hydrogen breath test — early rise in exhaled H_2 after lactulose or glucose indicates small bowel (not colonic) bacterial fermentation. **Treatment: rifaximin**, a non-absorbable antibiotic that reduces small bowel bacterial load.

ACTIVE LEARNING QUE

A patient with diabetic autonomic neuropathy develops bloating and diarrhea between meals. Which consequence of MMC absence best explains these symptoms?

- A. ○ Increased segmentation
- B. ○ Elevated motilin levels
- C. ○ Small intestinal bacterial overgrowth
- D. ○ Reduced gastric acid secretion
- E. ○ Accelerated gastric emptying

Explanation (requires correct answer)

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to LT, what are the key smooth muscle contraction patterns along the GI tract, and how do disruptions in coordination explain his constipation and bloating? LT's symptoms—intermittent constipation and postprandial bloating—point to impaired GI motility rather than a structural lesion.

His constipation likely reflects reduced phasic peristaltic activity in the colon, possibly due to diminished ICC pacemaker function or impaired ENS coordination, leading to slow tran disrupted MMC: without effective Phase III sweeps, residual content and gas accumulate, promoting bacterial fermentation and distension. Together, these findings illustrate how dysfunction in both phasic contraction coordination and interdigestive housekeeping can produce overlapping GI symptoms — a pattern consistent with conditions such as irritable bowel syndrome with constipation (IBS-C) or early diabetic enteric neuropathy.

Summary

- Tonic contractions are sustained and predominate at sphincters to maintain compartmental barriers, while phasic contractions are rhythmic and drive propulsion and mixing in the intestines.
- Interstitial cells of Cajal (ICC) generate slow waves that set the maximum frequency of phasic contractions; slow waves gate action potential firing but do not themselves cause contraction—unlike cardiac pacemaker potentials, which directly trigger each beat.
- Peristalsis uses ascending excitation and descending inhibition to propel content aborally, while segmentation produces non-directional mixing that maximizes mucosal contact and nutrient absorption.
- ICC dysfunction reduces slow wave frequency and disrupts spatial coordination of contractions, contributing to conditions such as gastroparesis and slow transit constipation.
- The migrating motor complex (MMC) cycles every 90–120 minutes during fasting; its Phase III activity front, driven by motilin, sweeps the small intestine clean of residual debris and bacteria.
- Absence of the MMC—due to motilin deficiency, autonomic neuropathy, or ENS disruption— rgrowth (SIBO), bloating, and malabsorption between meals.

Review Questions

1. Which feature best distinguishes tonic from phasic GI smooth muscle contractions?

- A. ○ Sustained contraction without relaxation
- B. ○ Rhythmic intermittent contractions
- C. ○ Driven by neural spikes only
- D. ○ Only in large intestine
- E. ○ Continuous rhythmic contraction

Explanation (requires correct answer)

2. How do ICC-generated slow waves differ from cardiac pacemaker potentials?

- A. ○ Generated by motor neurons
- B. ○ Identical to action potentials
- C. ○ Do not al
- D. ○ Originate in sinoatrial node
- E. ○ Absent in the duodenum

Explanation (requires correct answer)

3. Which contraction type slows transit and maximizes absorption in small intestine?

- A. ○ Peristalsis
- B. ○ Segmentation
- C. ○ Migrating motor complex
- D. ○ Tonic contraction
- E. ○ Slow wave generation

Explanation (requires correct answer)

4. What happens to GI motility when ICC pacemaker activity is disrupted?

- A. ○ Increased sphincter tone
- B. ○ Enhanced neurotransmission
- C. ○ Reduced frequency and coordination
- D. ○ Increased contraction frequency
- E. ○ Complete loss of slow waves

Explanation (requires correct answer)

5. What does absent migrating motor complex cause between meals?

- A. ○ Increased segmentation
- B. ○ Elevated motilin levels
- C. ○ Small intestinal bacterial overgrowth
- D. ○ Reduced gastric acid secretion
- E. ○ Accelerated gastric emptying

Explanation (requires correct answer)

H k?

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ **Distinguish tonic from phasic smooth muscle contractions, identify where each pattern predominates along the GI tract, and explain how sphincter tone regulates flow between compartments.**
02 ☐ **Explain how ICC-generated slow wave potentials set the basal contraction frequency of GI smooth muscle and gate action potential firing, and distinguish this from the mechanisms of skeletal muscle and cardiac pacemaker potentials.**
03 ☐ **Distinguish peristaltic from segmental contractions in terms of neural coordination, directionality, and functional role in propulsion vs. mixing.**
04 ☐ **Predict how disruption of ICC pacemaker activity alters the frequency and coordination of GI smooth muscle contractions.**
05 ☐ **Explain the phases and motilin-dependent regulation of the migrating motor complex, and predict the consequences of its absence on intestinal function between meals.**

Objective Confidence: 0/5